

Engagement Analytics & Strategic Programming Recommendations

Independent Consulting Project — Higher Education Sector

Tools & methods: Mixed-source data reconciliation, interval-based observational tracking, statistical comparison, executive presentation design | **Stakeholders:** Senior department leadership, full-time staff team

Situation

A university library department ran a weekly student engagement program with no structured way to evaluate performance beyond raw attendance counts. Existing records captured headcount but nothing about engagement quality, and no demand-validation had ever been cross-referenced against program design. Leadership needed a defensible, data-driven basis for restructuring the program ahead of the following semester, without the bandwidth to commission a formal analytics build.

Approach

I scoped, structured, and executed the engagement end-to-end. Rather than relying on the institution's single existing data source, I designed and ran a supplementary observational tracking system — interval-based engagement logging and individual visitor measurement — to capture a dimension the existing records couldn't: depth of engagement, not just attendance. I reconciled two structurally different data sources spanning a full semester, establishing explicit rules for when each should be trusted. Quantitative findings were cross-validated against an independent demand-preference dataset to confirm recommendations weren't artifacts of a single data source. The engagement closed with three deliverables built from one analytical base: a methodologically transparent strategy memo, a live self-maintaining tracking system handed off to staff, and an executive presentation delivered to senior leadership.

Scope

14 sessions analyzed · 2 independent data sources reconciled · 1 self-updating tracking system delivered · 3 strategic recommendations adopted

Skills Demonstrated

Designing original data collection where none existed · reconciling inconsistent data sources · validating findings against independent benchmarks · translating quantitative analysis into adopted strategic decisions · building tools for non-technical end users · independent ownership from scoping through executive delivery.

Specific findings and operational details have been omitted to respect the privacy of the partner organization. This summary reflects process and methodology, not internal performance data.